MolPallete — Corpus EDA

naveja_recap

/home/mogan/preprocessed/molpallete/coconut-flavordb_full_r040/naveja_recap

generated 2026-08-20 06:33

A · Provenance

field	value
sources	['flavordb', 'coconut']
method	naveja_recap
decomposition_params	{'ratio': 0.4, 'include_ring': True}
core_ratio	0.4
max_cores	10
max_rgroups	8
keep_stereo	True
neutralise	False
dedup	True
n_duplicates_removed	4412
size_filter	{'min_heavy_atoms': 5, 'max_heavy_atoms': 50}
graph_hash_version	2
rdkit_version	2026.03.2
created_at	2026-08-20T06:20:54

B · Composition

quantity	value
molecules	413,459
from coconut	391,965 (94.8%)
from flavordb	21,494 (5.2%)
decompositions	3,426,885
per molecule	mean 8.29, median 10, max 10
R-group slots	3,777,591
per decomposition	mean 1.10
k >= 2	8.39% of decompositions

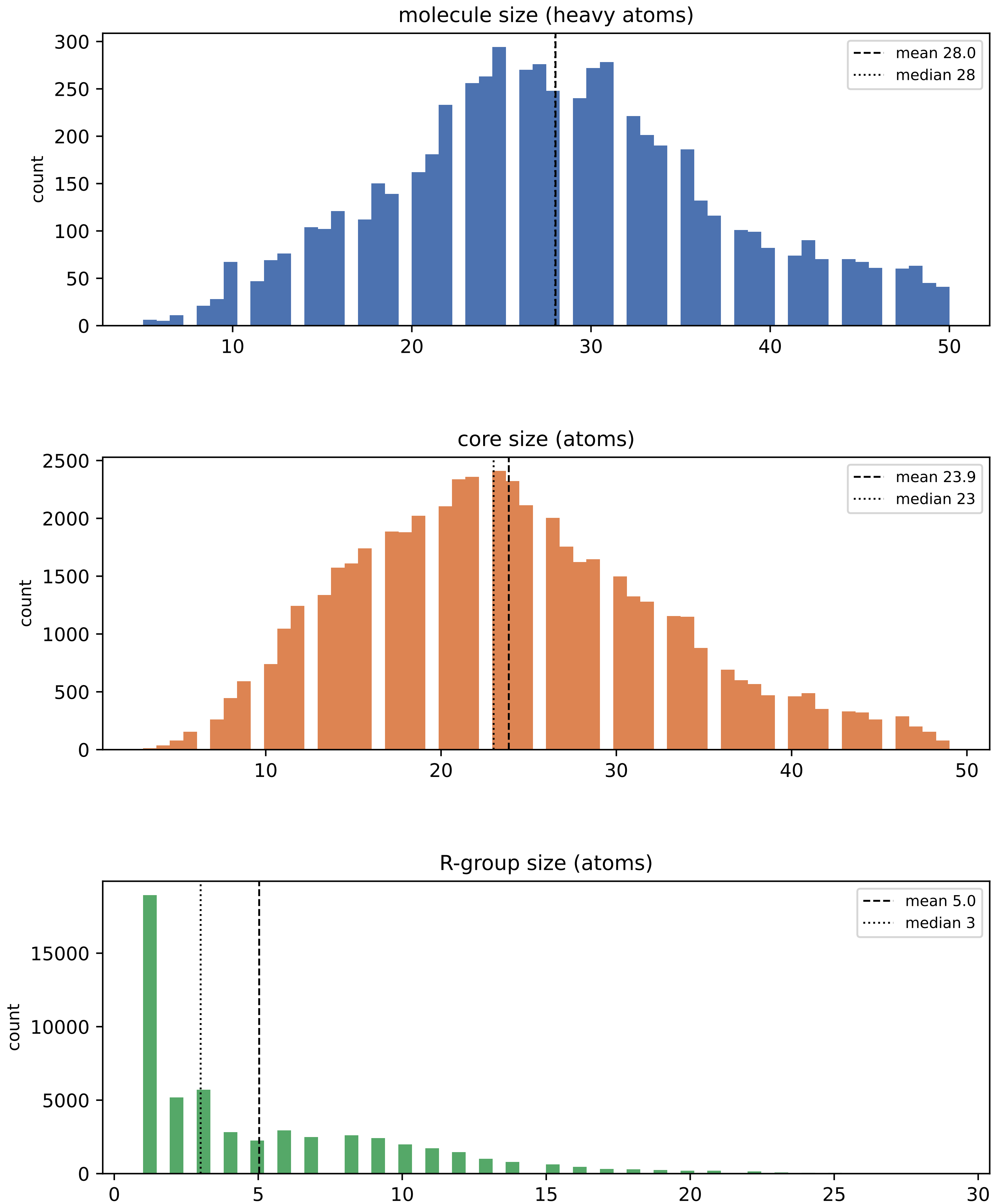
k>=2 is what the islinked subset space and the core-decoration objective have to work with.

C · Augmentation space

scheme	size
sampling_unit = molecule	413,459 instances/epoch
sampling_unit = decomposition	3,426,885 instances/epoch (8.3x)
full (mol, core, islinked) space	4,359,315 (10.5 per molecule)

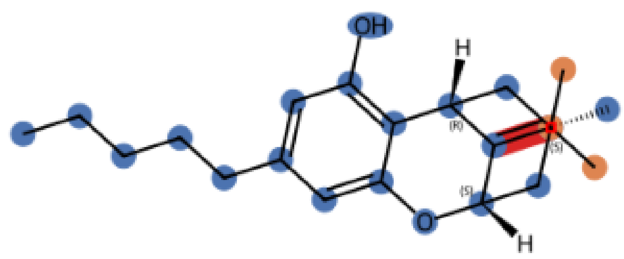
MolPLA's two stages: one molecule yields many cores; one core yields 2^k - 1 islinked subsets.

D · Size distributions

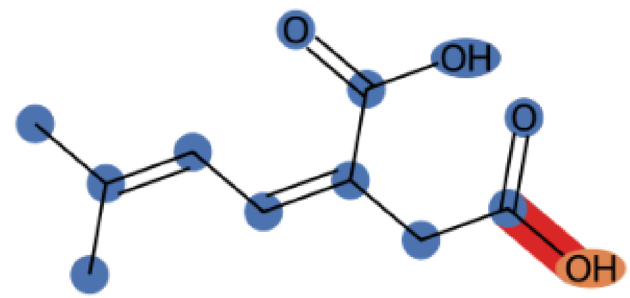


E · Example decompositions (ratio 0.4)

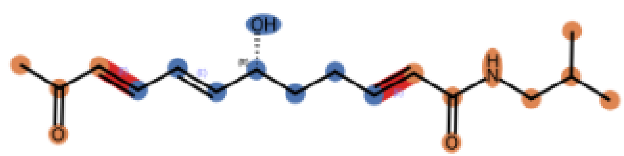
CNP0000074   k=1   core 20 atoms, R-groups [3]



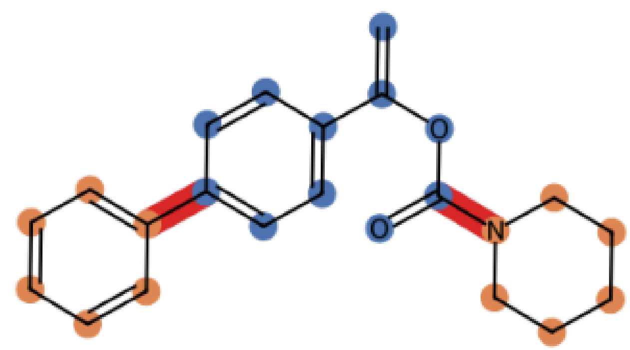
CNP0000161   k=1   core 12 atoms, R-groups [1]



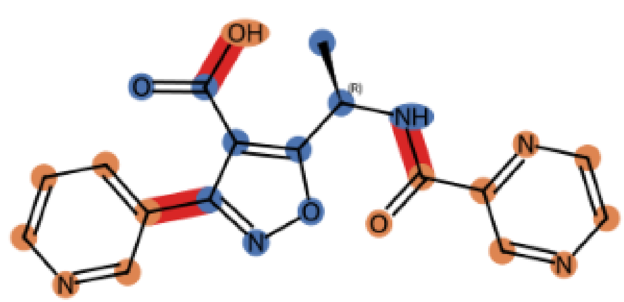
CNP0000307   k=2   core 8 atoms, R-groups [4, 8]



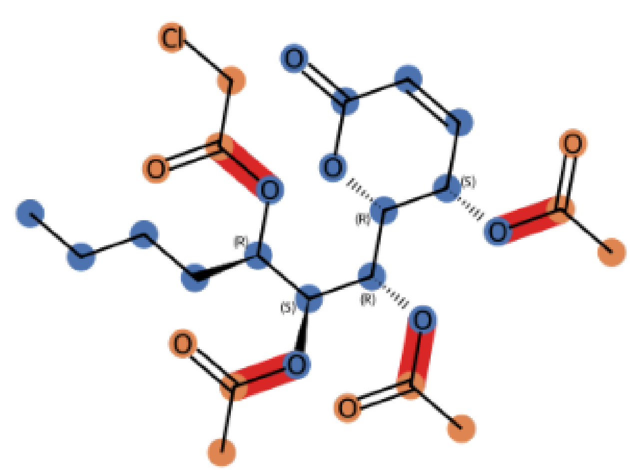
CNP0001292   k=2   core 11 atoms, R-groups [6, 6]



CNP0015426   k=3   core 10 atoms, R-groups [8, 6, 1]

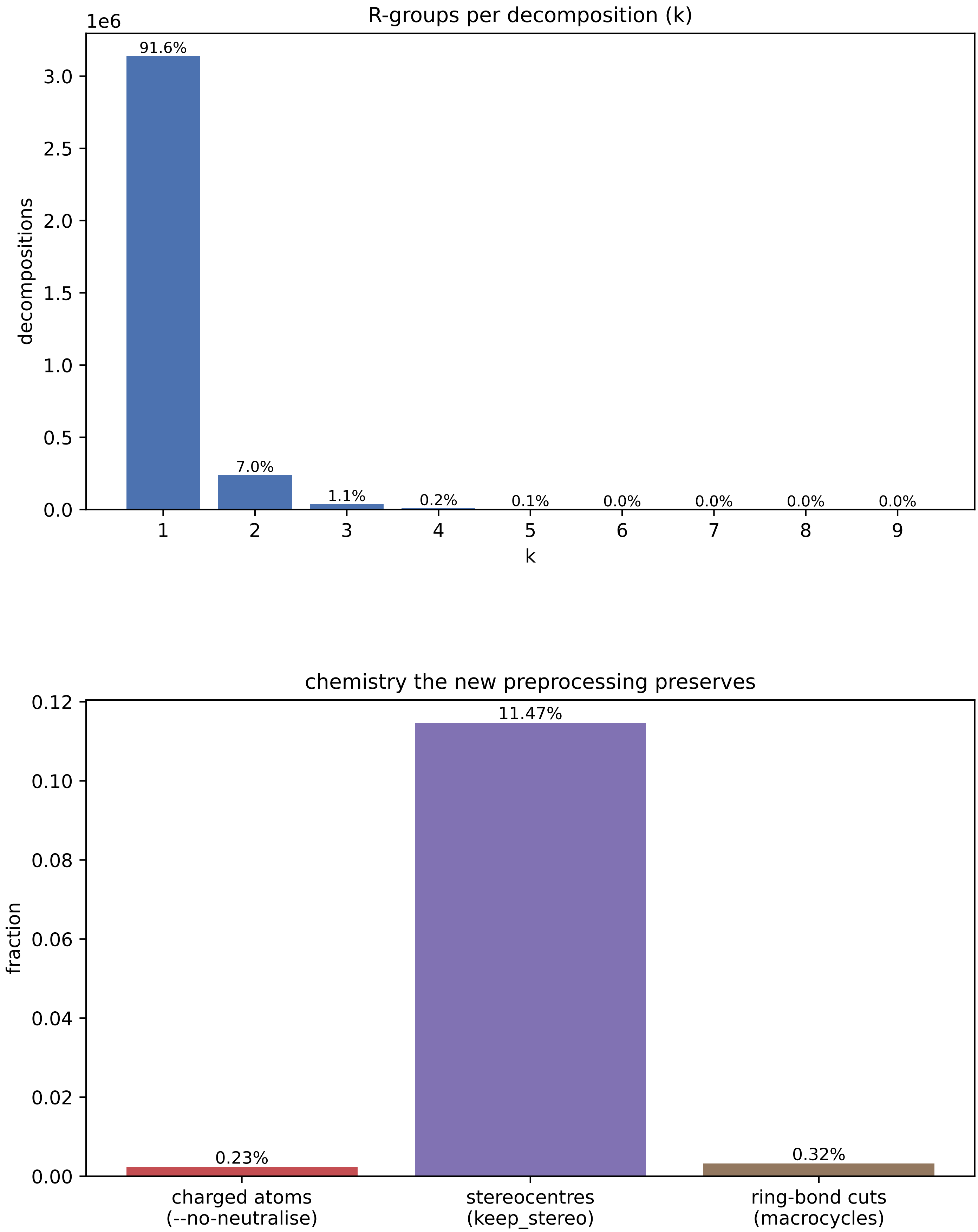


CNP0078333   k=4   core 18 atoms, R-groups [4, 3, 3, 3]

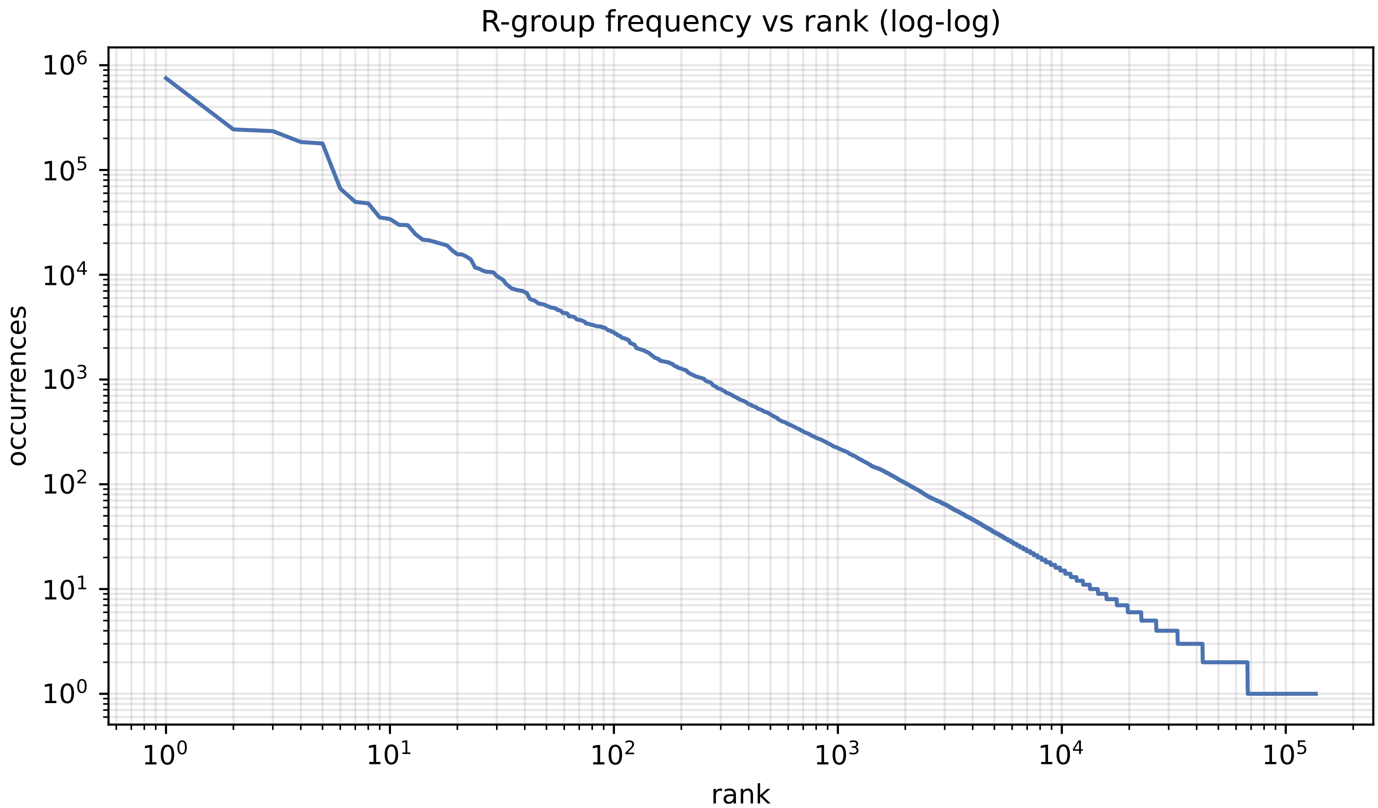


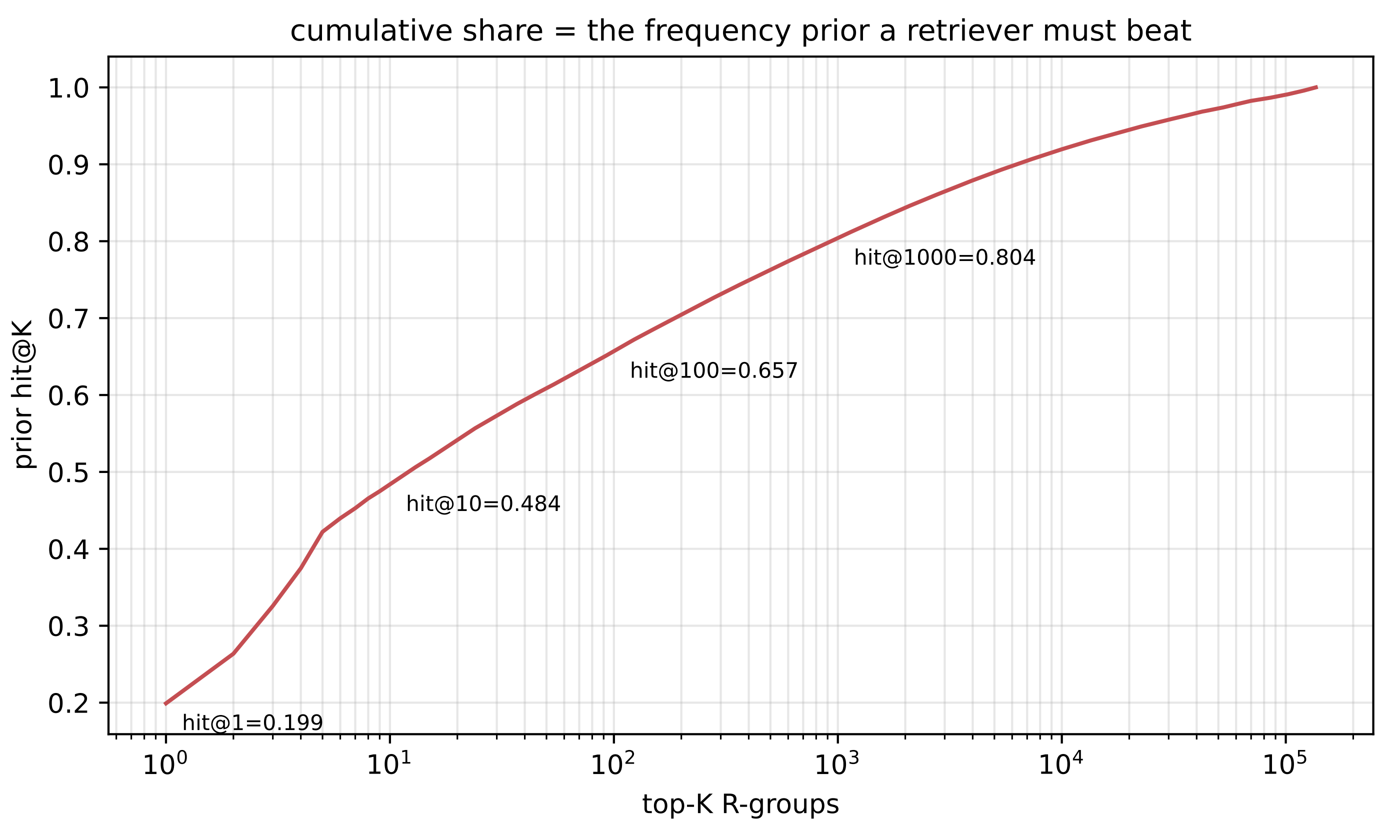
blue = core   orange = R-groups   red = cut bonds

F · Decomposition shape & preserved chemistry



G · R-group library skew





H · R-group library

quantity	value
distinct R-groups	136,159
occurrences	3,777,591
effective size (exp H)	364
top-1 share	19.91%
top-10 cover	48.4%
top-100 cover	65.7%

Effective size, not the raw count, is the number of classes retrieval actually chooses between.

I · Most frequent R-groups

count	share	SMILES
751,960	19.91%	*C
243,872	6.46%	*O
234,895	6.22%	*O
185,006	4.90%	*C(C)=O
178,856	4.73%	*OC
66,464	1.76%	*CC
49,580	1.31%	*CO
48,061	1.27%	*c1ccccc1
35,236	0.93%	*C(C)C
34,014	0.90%	*OC(C)=O
29,917	0.79%	*CCC
29,757	0.79%	*C(C)C
24,397	0.65%	*C(=O)O
21,606	0.57%	*CCCC
21,285	0.56%	*Cl
20,497	0.54%	*OC
19,911	0.53%	*c1ccc(OC)cc1
19,030	0.50%	*C(=O)OC
17,029	0.45%	*C(=O)c1ccccc1
15,681	0.42%	*OCC